Notes

Guesstimates

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How to Approach a Guesstimate (Step-by-Step):

- **Clarify the Question —** Ask clarifying questions: **Location, timeframe, units**

- **Break It Down —** Split into smaller logical steps using assumptions (top-down or bottom-up)
    - Use **top-down decomposition**:
        - Start with **Population** → Narrow to **Target users** → Narrow to **Usage behaviour** → Multiply.
        - Example frameworks:
            - **Population × Penetration × Frequency**
            - **# of Stores × Avg Customers per Store × Avg Spend**

- **Make Reasonable Assumptions —** Use population data, % penetration, frequency of use, etc.

- **Do the Math —** Keep calculations simple and organised

- **Sanity Check —** See if your final number seems reasonable

- **Communicate Confidence & Gaps**
    - Say: "This is a ballpark estimate; real data may differ depending on demographics and regulations."
    - Show that you understand **the drivers and limitations** of your model.

Popular Guesstimate Examples

1. Estimate the number of ATMs in India.
2. How many cups of coffee are consumed daily in Bengaluru?
3. How many flights land at Heathrow per day?
4. Estimate the annual revenue of Domino's in your city.
5. How many iPhones are sold in India per year?

Tips to Ace Guesstimates

- **Structure > Accuracy** → interviewer looks for *approach*, not the final number.
- Use **round numbers** (1000, 10%, etc.) to keep math easy.
- Break into **logical buckets** → population, penetration, frequency.
- Keep talking through your steps → show reasoning, not silence.
- End with a **sanity check** and possible real-world factors.

How to Choose Segmentation/Models in Guesstimates

*The **heart of guesstimates** is not the final number; it's whether you build a **logical, structured model**.*

1. Start with First Principles

- Ask yourself: *"What drives this number in real life?"*
- Example: *Pizzas sold per day* → driven by households ordering → frequency.
- Example: *Flights landing in Heathrow* → driven by runways × slots per day.
- Example: *iPhone sales* → driven by smartphone demand × Apple market share × seasonality.

Rule of Thumb: Choose the segmentation that best **mirrors reality**.

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2. Two Main Models

(A) Demand-side model (Population-based)

- Start from **population** → filter → usage frequency.
- Use when the question is **consumer behaviour driven** (food orders, taxis, phones).

Framework:

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Population → % Users → Frequency of Use → Period Conversion
```

(B) Supply-side model (Capacity-based)

- Start from **infrastructure** → supply limits → utilization.
- Use when the question is about **physical services/capacity** (flights, hotel rooms, ATMs).

Framework:

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#Units of Supply × Capacity per Unit × Utilization
```

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3. Segmentation Strategy

Break the big number into **MECE buckets** (Mutually Exclusive, Collectively Exhaustive):

- **By geography** (urban vs rural, Tier 1 vs Tier 2 cities)
- **By demographics** (age groups, income segments)
- **By behaviour** (high/medium/low frequency users)
- **By seasonality** (normal vs festive/holiday months)

Segmentation should help you:

- Make **assumptions more realistic**
- Avoid oversimplification (not all users behave the same)

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4. Choosing Which Segmentation

Ask: *"What variable creates the biggest differences in reality?"*

- Pizza orders → income groups / urban vs rural.
- Uber rides → working professionals vs students vs retirees.
- iPhone sales → income affordability tiers + festive months.
- Water bottles sold → climate zones + travellers vs residents.

If time is short → pick **1–2 high-impact segments** (not 10 tiny ones).

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5. General Steps (Reusable Logic)

1. Clarify scope
2. Choose model (demand-side vs supply-side)
3. Segment into 2–3 meaningful buckets (MECE)
4. State assumptions (round, easy numbers)
5. Do stepwise math
6. Sanity check
7. Provide a range, not an exact number

Example Mini-Case

Estimate the # of taxis in Delhi

- Demand-side: Delhi pop → % commuters → % who use taxis → frequency.
- Supply-side: Taxi utilisation → avg rides/day per taxi → total demand ÷ utilisation.
  Both converge → answer range.

So the **secret is not accuracy** — it's:

- Picking the **right driver(s)**
- Using **simple, logical segmentation**
- **Cross-checking** with another model

Mutually Exclusive, Collectively Exhaustive (MECE) Framework

"**Mutually Exclusive, Collectively Exhaustive (MECE)**" is a **problem-solving principle** often used in consulting, analytics, and structured thinking.

Mutually Exclusive (ME)

- **No overlaps** between categories.
- Each item should belong to only one category.
  Example: Classifying vehicles into **cars, bikes, buses, and trucks** — one vehicle can't fall into two categories at the same time.

Collectively Exhaustive (CE)

- **Covers all possibilities** without leaving gaps.
- Together, the categories must account for the entire universe of options.
  Example: If we only listed **cars and bikes**, buses and trucks would be missing → not CE.

Combined (MECE)

- Categories **don't overlap** (ME).
- Categories **cover everything** (CE).

Example: To classify people by age group:

- 0–12 (children), 13–19 (teenagers), 20–59 (adults), 60+ (seniors).
  This is **MECE** because:
- No person fits into two groups (Mutually Exclusive).
- Everyone fits into one group (Collectively Exhaustive).

Why MECE matters (esp. for Analysts/Consultants):

- Avoids duplication of effort.
- Ensures no blind spots.
- Makes communication structured and clear.

MECE on customer segmentation

Non-MECE Segmentation (overlaps + gaps)

- **High-income** customers
- **Young customers**
- **Urban customers**

Problem:

- A customer can be **high income AND young**, so overlap (not mutually exclusive).
- Doesn't cover all customers (what about rural or middle-income older customers?) → not exhaustive.

MECE Segmentation Example 1 (by age group)

- 0–17 years (Children/Teens)
- 18–34 years (Young Adults)
- 35–54 years (Middle-aged Adults)
- 55+ years (Seniors)

Mutually Exclusive: No customer can belong to 2 groups.
Collectively Exhaustive: Every customer falls in one group.

MECE Segmentation Example 2 (by spending behaviour)

- Price-sensitive shoppers
- Convenience-seeking shoppers
- Premium/loyalty shoppers

Mutually Exclusive: A shopper is classified into **only one dominant behaviour**.
Collectively Exhaustive: All types of customer shopping behaviours are covered.

MECE Segmentation Example 3 (for E-commerce)

- **New Customers** (first-time buyers)
- **Active Customers** (purchased in the last 90 days)
- **Churned Customers** (no purchase in the last 6 months)

Clear, non-overlapping states, and all customers fit somewhere.

Business Impact:
Using MECE segmentation helps:

- Design targeted marketing campaigns.
- Identify gaps in customer acquisition/retention.
- Avoid duplication (e.g., not sending multiple offers to the same customer group).

Ex:1 — How many pizzas are ordered in Mumbai per day?

1) Frame & approach

We'll use a **household-based model** (good for delivering food):

- Population → Households → Segment by income/ordering propensity → Orders per month → Convert to per day.
- Keep numbers round and state assumptions.

2) Assumptions (reasonable, rounded)

- Mumbai metro population **≈ 20 million**
- Avg household size **≈ 4** → **5 million households**
- Segment households by ordering behaviour:
    - **High-ordering (20%)**: **~6 orders/month**
    - **Mid (50%)**: **~2 orders/month**
    - **Low (30%)**: **~0.5 orders/month**
- Convert month → day using **÷ 30**

3) Math (step-by-step)

Households by segment:

- High: 20% of 5,000,000 = **1,000,000** HH
- Mid: 50% of 5,000,000 = **2,500,000** HH
- Low: 30% of 5,000,000 = **1,500,000** HH

Monthly orders:

- High: 1,000,000 × 6 = **6,000,000**
- Mid: 2,500,000 × 2 = **5,000,000**
- Low: 1,500,000 × 0.5 = **750,000**

Total per month = 6,000,000 + 5,000,000 + 750,000 = **11,750,000**

Per day ≈ 11,750,000 ÷ 30 = **391,666.7 ≈ 392k pizzas/day**

4) Sanity check (triangulation)

Store-capacity view: Suppose ~**2,000–2,500** pizza outlets (chains + local).
If each does ~**120–150** orders/day → **240k–375k/day**, same order of magnitude as **~392k/day**.
✔

5) Answer (with range)

~0.35–0.45 million pizzas/day, ballpark **~0.39M/day**.

What the interviewer looks for (how to narrate)

1. **Clarify scope**: "I'll estimate daily pizza orders, city = Mumbai metro, delivery + dine-in takeaways."
2. **Choose a model**: "I'll use a household model and cross-check with a supply-side (store) view."
3. **State assumptions** cleanly and round numbers.
4. **Show math** stepwise (no mental leaps).
5. **Sanity check** with a second method and give a **range**.
6. **Identify levers**: If any knob changes—household size, penetration, frequency—the result scales linearly.

Ex:2 — iPhones sold in India in Dec 2024

1) Clarify the Question

- Scope: Only **India**, only **iPhones**, only **December 2024** (1 month).
- Metric: **Units sold**, not revenue.
- Channels: Both **online + offline**.

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2) Choose an Approach

Top-down: population + penetration + replacement cycle

- Start from India's population → smartphone penetration → Apple market share → Dec seasonality spike.

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3) Assumptions (Rounded, Reasonable)

- India's population **≈ 1.4B**
- Smartphone penetration **≈ 60%** → **840M** smartphone users
- iOS share in India **≈ 5%** → **42M** iPhone users already
- iPhones typically replaced every **3 years (~36 months)** → annual demand **≈** 42M ÷ 3 = **14M iPhones/year**
- That's **≈ 1.2M iPhones/month on average**
- December is a **festive + holiday + year-end sales** month → **~2× normal month**

So **Dec 2024 sales ≈ 2.4M iPhones**.

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4) Final Answer

~2–2.5 million iPhones were likely sold in India in December 2024.
(A reasonable ballpark based on penetration, replacement cycle, and seasonality.)

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5) Narration in Interview

When answering, **walk them through this structure**:

1. *"I'll start from population, narrow to smartphone users, then Apple's market share, and factor in the replacement cycle."*
2. *"This gives me a baseline monthly sales, which I'll adjust upward for the December festive/holiday effect."*
3. *"Finally, I'll sanity-check using total smartphone shipments × Apple's share."*
4. *"Both approaches converge at ~2–2.5M, so that's my estimate."*

Ex:3 — Daily Coffee Cups in Bengaluru

Step 1: Clarify & Scope

- Bengaluru = city + surrounding metro population (~13 million people).
- Include: households + offices + cafés.
- Exclude: tourists (small %).
- Time period = **per day**.

Step 2: Choose a Model

This is a **demand-side problem** (depends on people's consumption habits).
So: **Population** → **Segment** → **Frequency** → **Cups/day**
We'll segment by **occupation/lifestyle**, since that drives coffee habits.

Step 3: Segmentation

Let's divide Bengaluru into 3 realistic consumer segments:

Segment	% of Population	Avg Cups/Day	Notes
Working Professionals (IT, corporate)	30%	2 cups	Office + morning brew
Students (college, school)	20%	1 cup	Mostly instant / café
Others (homemakers, retirees, etc.)	50%	0.5 cups	Occasional tea-coffee mix

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Step 4: Math

Population = 13 million

Segment	Population	Cups/Day	Total Cups/Day
Working Professionals	13M × 30% = 3.9M	2	7.8M
Students	13M × 20% = 2.6M	1	2.6M
Others	13M × 50% = 6.5M	0.5	3.25M

Total = 7.8 + 2.6 + 3.25 = 13.65 million cups/day

Step 5: Final Answer

~13–14 million cups of coffee are consumed daily in Bengaluru.

How to Explain in an Interview

> "I'd start from Bengaluru's 13M population, segment into working professionals, students, and others — since coffee habits vary strongly by lifestyle.
>
> Working professionals drink ~2/day (office + home), students ~1/day, and others ~0.5/day.
>
> That gives roughly 13–14M cups/day, which feels reasonable given the café culture here. I'd cross-check with café capacity and get a similar ballpark."

Ex:4 Total TVs Sold per Year in India

Step 1: Clarify & Scope

Before diving into numbers, clarify:

- Time frame: 1 year (latest year).
- Geography: India.
- Include: All TV types (LED, Smart, Non-smart).
- Include both urban + rural sales.
- Metric: **Number of TV units sold**, not revenue.

Step 2: Choose an Approach (Model)

For durable goods like TVs, there are **two logical approaches**:

(A) Replacement cycle model (very common for appliances)

→ Start from total households → TV penetration → replacement frequency → annual sales.

(B) New buyers + replacements

→ Sales come from *first-time buyers* + *old TV replacements*.
We'll use **Approach B**, since it gives a complete view.

Step 3: Build Assumptions

Population and households

- India's population **≈ 1.4 billion**
- Avg household size **≈ 4.5** → **~310 million households**

TV ownership (penetration rate)

- Urban households: ~85% have a TV
- Rural households: ~50% have a TV
- Urban: Rural split ≈ 35%: 65%
  → Weighted penetration = (0.35×85%) + (0.65×50%) = ~63%

Households with a TV = 310M × 63% = **~195M households**

Replacement cycle

- TVs last about **8 years on average** before replacement.

  → Annual replacement demand = 195M ÷ 8 = **~24M units/year**

New buyer demand

- Households without TV = 310M − 195M = 115M
- Assume ~5% of these buy their **first TV each year** → 115M × 5% = **~6M units/year**

Step 4: Total Annual Sales

Total = Replacement + New buyers
= 24M + 6M = **~30 million TVs/year**

Step 5: Sanity Check (Alternate Lens)

- India's consumer durable market data roughly supports 25–35M TVs/year (reasonable).
- Urban sales spike during festive seasons (Diwali, Onam, etc.) → our number aligns well. Looks realistic and justifiable.

Step 6: Final Answer

≈ 30 million TVs are sold per year in India.

How to Explain in an Interview

"I'd approach this via a household-based model.
India has around 310M households, ~63% penetration, and TVs are replaced every 8 years — giving ~24M replacement sales. Add ~6M first-time buyers annually, and you get about 30M TVs sold per year.

This aligns with industry estimates, so it's a reasonable ballpark."

Bonus: Alternate Models (for flexibility)

1. **Income-segmentation model**
    - High-income (15%) → buy every 5 years
    - Mid-income (50%) → every 8 years
    - Low-income (35%) → every 10 years
    - Weighted replacement rate → similar to ~30M total

2. **Retail-side check**
    - Suppose 75k electronics stores × 100 TVs/month × 12 = 90M (overestimate).
    - But if only 1/3rd are major sellers → ~30M — consistent

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